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HighGen—Comparative genomics resources for cannabis

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UCSC's chain/net tools and MULTIZ are tuned for cross-species vertebrate panels. With **HighGen**, we adapted them in Galaxy across **217** ***Cannabis sativa*** **assemblies** anchored on `cs10` through standardized repeat masking, a held-out test of an empirical substitution matrix, chunked extension, and streaming 64-bit dynamic programming. We pair the alignment with 43-genome Panagram *k*-mer bitmaps, a TOGA2 orthology pilot, and a 78-family phenotype ledger.

Galaxy PROJECT

217-assembly Hub
Attach genome.ucsc.edu/hubspace/74/richard%252Dburhans/highgen_cannabis/hub.txt in the UCSC Genome Browser (Track Hubs → My Hubs).

A Four cross-species bottlenecks

In *Cannabis sativa*, interspersed repeats (63 to >70%, mostly LTR retrotransposons) surround dense tandem metabolic arrays (Grassa et al. 2021, Lynch et al. 2025, Pike et al. 2026). Projecting 216 query assemblies onto the linear `cs10` (CBDRx) reference coordinate grid confronts four vertebrate-default obstacles:

1 Masking bias
Arrived soft-masking spans 0.00–80.86% across 218 assemblies, and KegAlign (Gulhan et al. 2025) seeds only through uppercase bases

2 Cross-species matrix
HOXD70 (Chiaromonte et al. 2002) was trained below 70% identity, where 49.7% of pooled *Cannabis* pairs fall (Ts/Tv = 0.872 vs. a 0.74 chance level)

3 Superlinear scaling
Whole-chr LASTZ (Harris 2007) scales as $O(n^{1.9})$, while 217-way MULTIZ (Blanchette et al. 2004) overflows 32-bit scores and buffers MAFs in RAM

4 Macro-syntenic nets
Human-mouse `netFilter -syn` (Kent et al. 2003, `maxFar = 200 kb`) prunes rearranged tandem gene clusters

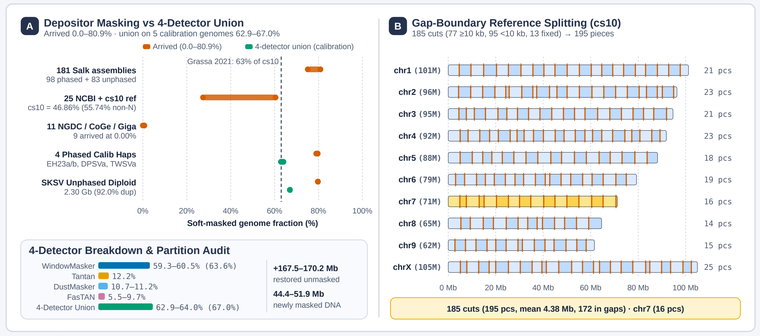


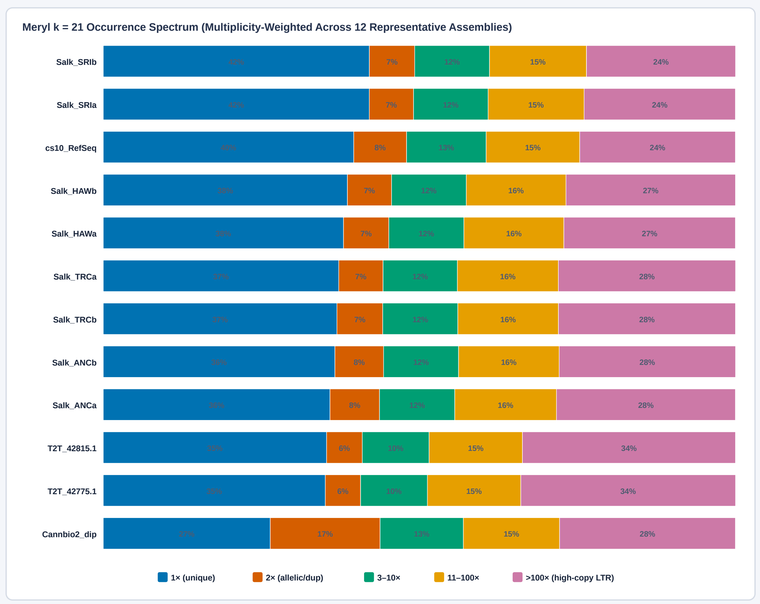
Figure 1 | Standardized masking and gap-boundary reference splitting. Panel **A**: arrived masks (0.0–80.9%, 218 assemblies) vs. the 4-detector union on 5 calibration genomes (62.9–67.0%), line at Grassa et al.'s (2021) 63% for `cs10`. Panel **B**: across `cs10`, 185 chromosomal cuts (77 at N-runs ≥ 10 kb, 95 at N-runs under 10 kb, and 13 fixed 5 Mb cuts in `wfi_cs10_splits.bed`) partition the assembly into 195 ~5 Mb pieces plus 26 unplaced scaffolds, mitochondrion `NC_029855.1`, and `small.00`. `mafSplit` divides straddling blocks without base loss, and `maf_concat` rejoins the pieces.

B Standardized 4-detector repeat masking

We curated 221 assembly records—217 non-redundant in MULTIZ (`cs10` plus 216 queries across 164 prefix groups and 156 plants, scored with `rosales_odb12` BUSCO, Lynch et al. 2025, Pike et al. 2025, 2026). To replace incompatible depositor masks, our `brc-tools` UDT wrappers strip arrived masks and unite four *de novo* detectors (Morgulis et al. 2006a, 2006b, Frith 2011, Myers 2026):

`fasta-uppercase` → **WindowMasker** **DustMasker** **Tantan** **FasTAN** → **bedtools merge**

Assembly / Cohort	Arrived Mask	4-Detector Union	Partition Audit and Compute
181 Salk assemblies (98 phased + 83 unphased)	75.12–80.86%	**62.19–67.04%** (5-genome pilot: 22.61 → 4.85 pt spread)	Depositor-only masking frees ~22% of each calibration haplotype to seed (repeat status unmeasured)
25 NCBI GenBank + `cs10`	27.42–60.43% (`cs10` 46.86% total, 55.74% non-N)		Leaves 44.26% of `cs10` non-N bases open to seeding
11 NGDC / CoGe / GigaDB	0.00–0.68% (9 kept at 0.00%)		Unmasked queries seed across all of their bases
4 Phased Calibration Haps (`EH23a/b`, `DPSVa`, `TWSVa`)	**79.00–79.61%**	**62.90–64.02%** **(1.12-pt window)**	**WM 59.30–60.53%, Dust 10.7–11.2%, Tantan 12.2%, FasTAN 5.5–9.7% · 3.64–3.99 CPU-h/hap**
Diploid `SKSV` (2.30 Gb, 92.0% BUSCO dup)	79.71%	**66.95%** (4.05-pt panel window, WM 63.60%)	432.1 Mb restored, 139.1 Mb newly masked · 10.34 CPU-h



12-genome Meryl *k* = 21 occurrence spectrum: Phased haplotypes (`cs10`, 8 Pike et al. 2026 ONT, 2 IMPD T2T) carry 35–42% `1×` and 6–8% `2×` multiplicity-weighted *k*-mers, whereas uncollapsed diploid `Cannbio-2` (Braich et al. 2020) carries 27% `1×` and 17% `2×`.

Across phased haplotypes, the union restores 167.5–170.2 Mb of depositor-masked DNA while newly masking 44.4–51.9 Mb—a net unmasked gain of +115.9 to 123.2 Mb per haplotype with exact lowercase FASTA ↔ 0-based BED parity.

C Empirical intra-species substitution matrix

We adapted the Chiaromonte et al. (2002) log-odds protocol across two phased haplotype pairs (`EH23a × EH23b` and `DPSVa × TWSVa`, 95.3% `rosales_odb12` BUSCO completeness, 1.8–2.0% duplication). We restored her CDS excision (–7.3% of chained segments) and single-chain colinearity (`--max-chains 1`), and inverted the training window to `[85, 95)%` identity, 73.9% of whose bases are cross-chromosome. The filter enriched same-chromosome over cross-chromosome matches **8.2-fold** on a `Cb2 × cs10` slice (54.5% vs. a 6.6% median) and **10.9-fold** over all 200 `EH23a × EH23b` groups (8.08% vs. 0.74%).

331.05M
Pooled `[85, 95)%` pairs, symmetrized log-odds (`Ts/Tv = 2.119`, $H = 1.4140$ bits, either pair alone within 1 unit)

0/10 vs. human-chimp
UCSC's human-chimp matrix beat ours on all ten held-out 20 Mb arms (9/10 by minimap2). Ours beat HOXD70 10/10 by colinearity, 3/10 by minimap2

5,033×
`axtChain` shuffled-null suppression (`EH23a chr1 × Morus`) + scaled `682/51` gaps (70.6% single coverage)

We normalized scores to a maximum match of 100 (`+80` A/T, `+100` C/G, `−92` transitions, `−191` A-C/G-T, `−197` C-G, `−198` A-T transversions). Across 40 false-match budgets (colinearity referee, gap-free), the matrix beat unit `±100` on **10/10** at 20 Mb, **10/10 loci**, and **62/70 windows**. Its gain over the unchained matrix was small (0.02–0.11% at 20 Mb) and 6/10 under minimap2. Retraining at 97–99.5% narrowed the human-chimp gap without closing it. Production used ours.

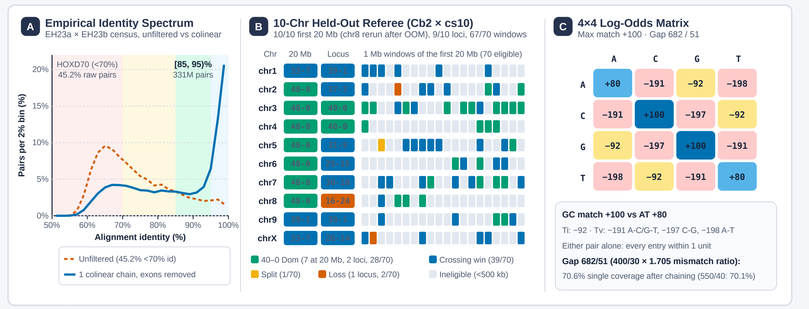


Figure 2 | Intra-species matrix calibration and 10-chromosome × 20-window held-out referee. Panel **A**: empirical identity spectra and single-chain colinearity separation. Panel **B**: colinearity-referee wins vs. HOXD70 on the first 20 Mb of 10 held-out `Cb2 × cs10` chromosomes and 70 1 Mb windows. Panel **C**: symmetric 4×4 log-odds weights with `682/51` gap scaling.

D Collection-native LASTZ and MULTIZ 11.3/11.4

- **Growler LASTZ (`growler_ucsc_pair`, LASTZ 1.04.52/1.04.60):** Couples KegAlign v0.3.4 GPU collection seeding (Gulhan et al. 2025) with indexed `.2bit` archives, compact LAV, and piped PSL chaining (Harris 2007, Kent et al. 2003). Batching anchor files into `≤50,000`-segment chunks cuts `EH23a × EH23b` gapped extension **7.8-fold** (`389.0 → 49.8 slot-h`, `1.79 h` for 100 jobs, 18% above `batched_lastz`, median 28 min/job, under `3.97 GB` RAM). `growler_lastz_lav`, which chunks identically, reproduced **99.99%** of UCSC's `hg38 × panTro6 chr21` `all.chain` and `rbest.chain` pairs, and `doRecipBest` chains on one query per class.
- **Storage and 218-run production cohort:** Purging transient intermediate collections after workflow completion cut total Galaxy history storage across all 218 runs (54,554 jobs, 10,325 job-hours) from **14.96 TB to 6.50 TB (–56.5%,** `8.45 TB` saved, summed). Syntenic nets cover a median **65.2%** of `cs10` non-N bases across 216 queries (range 12.2–85.1%, vs. **70.0%** and 29.3–88.1% in `rbest`, with 40.0% diploid vs. 54.0% phased median query-side coverage).
- **Streaming 64-bit column-count MULTIZ 11.3/11.4:** In v11.3 (`multiz#19`, Blanchette et al. 2004), coordinate-ordered streaming cursors replaced in-memory MAF buffering (v11.2) and `int64_t` accumulators fixed 32-bit overflow across 29 deep blocks on `NC_044376.1.02` (24 with scores $\geq 2^{31}$). Peak memory fell **51.7-fold** on `hg38` 5-way `chrX` (`677 → 13 MiB`). In v11.4 (`multiz#28`), factoring `mz_yama.c` DP loops through column nucleotide counts (c_A, c_C, c_G, c_T, c_N) sped up one upstream-timed deep merge **33.8-fold** (`981 → 29 s`). On one host running no other alignment job, a whole 4.4 Mb piece took `3.9 → 0.80 h` (**4.9-fold**, medians of three interleaved runs per release). Output is identical apart from `#` lines.

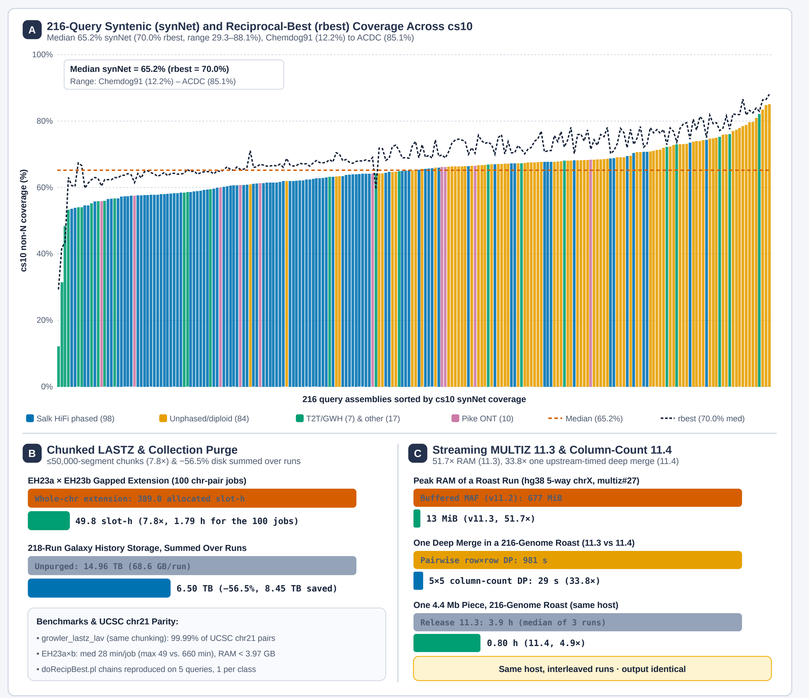


Figure 3 | 216-query syntenic coverage waterfall and scalability benchmarks. Panel **A**: sorted `synNet` (median 65.2%, range 12.2–85.1%) and `rbest` 1-to-1 (median 70.0%, range 29.3–88.1%) coverage across 216 queries by assembly class. Panel **B**: chunked LASTZ speedup, `hg38 chr21` parity, and 218-run storage purge. Panel **C**: MULTIZ v11.3/v11.4 streaming RAM reduction (51.7×, human 5-way) and column-count DP speedup (33.8×, one merge).

E Tandem metabolic clusters and syntenic net limits

- **`cs10` tandem census and permutation audit:** Adapting MCScanX's tandem criteria (Wang et al. 2012) with BLASTp $E \leq 10^{-10}$ and the top 10 hits, **3,035** of `cs10`'s 25,296 RefSeq protein-coding genes lie in 1,208 tandem arrays ($K = 0$) and **5,123 (20.3%)** in 1,796 clusters ($K \leq 10$). These clusters include **41 terpene synthases (TPS)** (31 in 9 clusters + 10 singletons) and **35 cannabinoid oxidocyclases (BBE-like)** (29 in 9 clusters + 6 singletons including `CBDAS`). Across 20,000 cluster-level permutations, nominal TPS low-identity enrichment (37.0% vs. 11.0% controls, $p = 0.022$) does not survive family-wise correction ($p_{FWER} = 0.12$, threshold-free $p = 0.40$, BBE-like 2.8%, $p = 0.59$).
- **Functional `CBDAS` vs. B-locus pseudogenes:** At `cs10` functional `CBDAS` (`LOC115697762`, chr7:30.98 Mb, Grassa et al. 2021), our 217-way alignment separates **79 assemblies** (59 plants, including 56 Lynch full-length `CBDAS`) at `≥98.5%` identity and **6 assemblies** below `93.0%` (`≥50%` CDS) from **77 assemblies** (67 plants) at `93.0–98.5%`. In that band, 63 carry Lynch et al. (2025) annotations placing the alignment on the ~94% B-locus pseudogene that Laverty et al. (2019) described (62 in THC-dominant cultivars), and 14 carry none.
- **`netFilter -syn` pruning vs. paralog reuse:** Applied beyond UCSC's contiguity rule, its human-mouse cutoffs (`maxFar = 200 kb`, plus `minTopScore = 300,000` for 14.5% of losses) pruned **4.7% of TPS** (418/8,856: 4.2% clustered, 6.4% singleton) and **10.7% of BBE-like** (807/7,560: 7.5% clustered, 26.2% singleton vs. 5.2%/13.1% controls). In `synNet`, this dropped 54 `CBDAS` assemblies (under `50%` CDS), including Finola's intact 99.4% transposed copy (`CM011610.1:21.84 Mb`, 26.8 Mb from its enclosing gap). Removing `netFilter -syn` leaves **46.4%** of recovered pairs (569/1,225) overlapping another family gene's interval. Resolving them needs `rbest` nets or recalibrated thresholds with duplication-aware orthology (TOAST/BOAST, Hou et al. 2006, 2009).

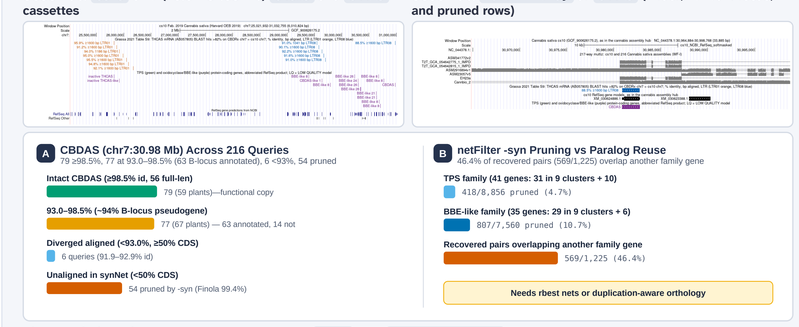


Figure 4 | Live HighGen UCSC Track Hub on `chr7` and `netFilter -syn` trade-off. Browser views of the 6 Mb cannabinoid synthase super-cluster (left) and `multiz217way` at `CBDAS` (right). Below: (**A**) 216-query `CBDAS` stratification and (**B**) `netFilter -syn` pruning vs. the 46.4% of recovered pairs overlapping another family gene.

F Panagram bitmaps, TOGA2 pilot, and phenotype ledger

- **Panagram v1.1.0 pan-*k*-mer indexing (`panagram_index` / `panagram_view`, Jenike et al. 2026, Kokot et al. 2017, Aylward et al. 2023):** Tier P1 (17 genomes: `cs10` + 16 Salk haplotypes from 9 plants, 13.3 Gb) selected $k = 21$ over $k = 31$ on higher 1 Mb window Pearson r with `rbest` identity across **16/16 Salk haplotypes** (mean $r = 0.7150$ vs. 0.6901, $\Delta r = +0.0249$ within the 0.05 equivalence band). Tier P2 indexed **43 genomes** (32.64 Gb, 5.05 h on 16 Laila slots, 1.14 GB index). It passed $r \geq 0.50$ in **41 of 42 chain-scored queries** ($r = 0.5848$–0.8118, `Pink pepper` $r = 0.7376$) and separated non-PAR `chrY` cleanly (3/3 males 0.976–0.987 vs. 3/3 V3 females ≤ 0.0773 across 205,619 subsampled `AH3Ma` -absent positions). It flagged 2 Guppy 5 SUP trio-binned ONT haplotypes on `G3` count tolerance (`Punto Rojo` –10.4%, `Cherry Pie` –10.8%, Pike et al. 2025). Only one Sri Lankan ONT haplotype fell below 0.50 (`ASM5191173v1`, $r = 0.4763$, Merqury QV 24.01/24.11, Pike et al. 2026), for reasons still under investigation.
- **TOGA2 v2.0.9 orthology pilot (Kirilenko et al. 2023, Malovichko et al. 2026):** On `cs10`, we stratified all **25,261** nuclear protein-coding genes (33,639 transcripts, excluding mitochondrion `NC_029855.1`) into **24,503 standard** and **758 reference-anomaly** genes alongside 221 inventoried `all.chain` datasets (1.45 TB). On its zero-divergence identity-chain control, TOGA2 called **25 of 24,503 lost/uncertain (0.10%)**—1 L + 24 UL against a ≤ 0.5% bar. On `EH23a`, the production, loose, and `RepeatFiller`-filled-and-cleaned chains called **2,077 (8.48%)**, **1,778 (7.26%)**, and **1,660 (6.77%)** standard genes L/UL. Filled and production arms disagree on 713 of 2,161 L/UL/M genes (33.0%), while production yields the fewest multi-copy false presences (4,306 standard genes vs. 5,058 loose and 5,245 filled). Because our top-ranked native-model check (`B6`) is still pending, the pilot selects no arm and reports no gene loss.
- **Pedigree-verified phenotype ledger:** We joined 40 sources across 215 assemblies—30 publications (29 papers, including 2 supplements, plus 988 Lynch et al. 2025 Figshare rows across 190 assemblies from 43 audited extractions in `passed.tsv`) and 10 NCBI BioSamples. This yielded **1,341 trait records** across **154 plants in 78 pedigree families** (69 plants in the largest Salk family, 71 singletons). Under strict `measured` / `reported` filtering, 0 of 14 traits pass our ≥30-family, ≥2-class (≥5 families/class) association screen. `Varin` spans 30 families (37/38 non-WT plants in `HO40` across 3 classes), while `Chemotype` reaches 30 families with 2 classes ≥5 only if Lynch's HPLC methods sentence covers its table column.

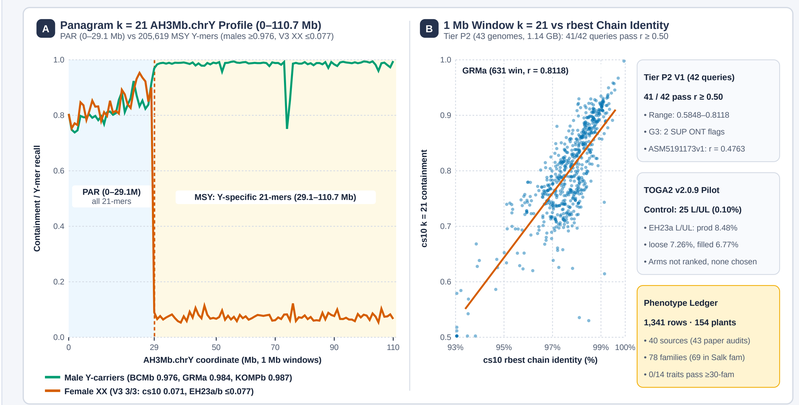


Figure 5 | Alignment-free Panagram `chrY` profile and `rbest` correlation. Panel **A**: chromosome-wide `AH3Mb.chrY` (0–110.7 Mb) $k = 21$ profile of PAR all-21-mer containment (0–29.08 Mb) and MSY male-specific 21-mer recall across 205,619 subsampled positions. Panel **B**: all 631 eligible 1 Mb windows on `GRMa` ($r = 0.8118$) with Tier P2, TOGA2, and phenotype scorecards.

Public tools, workflows, and 228-stanza (219-assembly) browser hub

4-detector masking → `brc-tools` **GPU seeding + chunked LASTZ →** `growler_lastz_lav` **217-way streaming TBA →** `MULTIZ 11.3/11.4` **Pan-*k*-mer + orthology →** `Panagram` **and** `TOGA2`

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HighGen Hub · Galaxy Tools · Repository